

Jingjie WANG

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RESEARCH INTERESTS

My research centers on human factors and ergonomics in complex, safety-critical systems, with applications in automated transportation, smart mobility, healthcare, and evacuation systems. I integrate behavioral experiments, computational modeling, and data-driven analysis to understand human decision-making and human-AI interaction, so as to design safer, more adaptive, and human-centered systems.

EDUCATION

Purdue University | Edwardson School of Industrial Engineering

Ph.D. in Industrial Engineering, advised by Dr. Brandon Pitts

West Lafayette, USA

Aug 2025-May 2029 (Expected)

- **GPA:** 4.00/4.00

Sichuan University | Pittsburgh Institute (SCUPI)

B.S. in Industrial Engineering, advised by Dr. Xiaomei Tan

Chengdu, China

Sep 2021-Jun 2025

- **GPA:** 3.93/4.00
- **Thesis:** Impact of Driver Experience on Takeover Performance during Silent Automation Failure

RESEARCH EXPERIENCE

Next-generation Human-systems and Cognitive Engineering (NHanCE) Lab, Purdue University

Aug 2025-Present

Graduate Research Assistant (PI: Dr. Brandon Pitts)

- Lead human factors research on human-AI interaction in safety-critical systems, including automated vehicle-initiated takeovers, large language model applications in driving, and human-vehicle teaming. Design simulator and survey studies and analyze behavioral, eye-tracking, physiological, and self-reported data to examine trust, transparency, explainability, decision-making, and human-AI teaming.

Pedestrian Evacuation and Intelligent Emergency Response Lab, Sichuan University

Sep 2023-Jun 2025

Undergraduate Research Assistant (PI: Dr. Wei Xie)

- Conducted computational and data-driven research on pedestrian dynamics, emergency evacuation, and pedestrian-non-motorized vehicle interactions. Developed agent-based and social-force models to evaluate assistance behavior and gait dynamics and analyzed naturalistic trajectory data to characterize individual and group movement, collision avoidance, and safety risks.

Human-centered AI (HAI) Lab, Sichuan University

Sep 2022-Jun 2025

Undergraduate Research Assistant (PIs: Drs. Xiaomei Tan & Rong Yin)

- Contributed interdisciplinary human factors research across automated driving, healthcare, and immersive learning. Led an undergraduate thesis on driver takeover performance during silent failures and contributed to projects evaluating automated error-detection systems for laparoscopic training and VR-based educational technologies, using experimental design, usability evaluation, behavioral data analysis, and statistical methods.

HONORS & AWARDS (SELECTED)

Sichuan University Ten Outstanding Undergraduate Students Finalist (Highest Undergraduate Honor, Top 1%)	2025
Outstanding Graduate of Sichuan Province (Top 5%) and Outstanding Graduate of Sichuan University (Top 5%)	2025
Outstanding Teaching Assistant Award in Sichuan University-Pittsburgh Institute (Top 10%)	2025
Sichuan University Special-Class Scholarship (¥5,000, Highest Undergraduate Scholarship, Top 1%)	2024
Sichuan University-Pittsburgh Institute Dean's List (¥5,000, Top 5%)	2024
Finalist in COMAP Interdisciplinary Contest in Modeling (ICM) (Team Captain, Top 2% worldwide)	2024
Sichuan University Comprehensive Scholarship (Top 5%)	2022/2023
Sichuan University Outstanding Student Leader (Top 10%)	2022/2023/2024

PUBLICATIONS (†Co-first authors; *Corresponding author)

Peer-Reviewed Journal Papers

- [J5] Wang, J., Luo, Y., Zhou, X., Zhao, W., Lei, Y., & Xie, W. (2025). Impact of assistance behavior on evacuation efficiency in high-rise hospital inpatient buildings: An agent-based simulation study. *Journal of Building Engineering*, 112, 113780. <https://doi.org/10.1016/j.jobe.2025.113780>
- [J4] Luo, Y., Wang, J., Yan, Z., He, J., Fu, L., Wang, S., Han, Y., Fu, Y., Wang, X., Li, K., Yin, R. & Pu, D. (2025). Enhancing laparoscopic surgery training: a comparative study of traditional models and automated error detection system. *BMC Medical Education*, 25(1), 1-12. <https://doi.org/10.1186/s12909-025-07242-3>

- [J3] Li, Z., Xie, Y., **Wang, J.**, Huang, G., Yu, L., Zhang, K., Li, J., & Liu, C. (2026). Wampee-YOLO: A High-Precision Detection Model for Dense Clustered Wampee in Natural Orchard Scenario. *Horticulturae*, 12(2), 232. <https://doi.org/10.3390/horticulturae12020232>
- [J2] Xie, W., Xuan, C., **Wang, J.**, Ming Lee, E. W., & Tian, B. (2026). A novel gait-based bipedal social force model for simulating pedestrian dynamics on inclined surfaces. *Safety Science*, 197, 107119. <https://doi.org/10.1016/j.ssci.2026.107119>
- [J1] Hu, B., Huang, L., Chang, H., Ji, Z., Yan, Z., Qu, D., **Wang, J.**, Qu, F., & Liang, H. (2025). A review of key parameters affecting inorganic scaling in thermal, pressure, and osmosis-driven membranes for produced water desalination. *Separation and Purification Technology*, 354(4), 129023. <https://doi.org/10.1016/j.seppur.2024.129023>

Peer-Reviewed Conference Proceedings

- [C1] Meng, Y.[†], **Wang, J.[†]**, Xiao, T., Liu, Z., Liu, B., Long, Q., Liu, J., & Gao, N. (2026). Evolutionary games: Use VR games to enhance users' understanding of evolution and cultivate interest in biological knowledge. In *Proceedings of the Augmented Humans International Conference 2026* (pp. 307-324). Association for Computing Machinery. <https://doi.org/10.1145/3795011.3795033>

Research Works in Progress

- [W8] **Wang, J.**, Wang, Q., Li, G., & Pitts, B. J. Tool, Teammate, or Rival? A Systematic Review of Evolving Human-AI Roles in Agentic Transportation. Under review at *Frontiers in Computer Science*
- [W7] **Wang, J.**, Aguirre Garrido, V., Nguyen, T., Marion, J., Yih, Y., & Pitts, B. J. One Size Does Not Fit All: How AI Is Used for Emotional Support across Age Groups. Under review at *International Journal of Human-Computer Interaction*
- [W6] **Wang, J.[†]**, Xuan, C.[†], Lee, E. W. M., & Xie, W. Pedestrian-unit structure and spatial negotiation with non-motorized vehicles: A drone-based trajectory study at a signalized crosswalk. Under review at *Accident Analysis & Prevention*
- [W5] **Wang, J.***, Nguyen, T., Marion, J., Yih, Y., & Pitts, B. J. AI Tools for Emotional Support: An Initial Survey of Age-Related Differences in Perceptions and Use. Accepted at *Proceedings of the Human Factors and Ergonomics Society Annual Meeting, 2026*
- [W4] **Wang, J.***, & Pitts, B. J. From Code to Collaboration: A Cognitive Agent Framework for Large Language Model (LLM)-Based Human-Vehicle Teaming. Proceedings of the Human Factors and Ergonomics Society Annual Meeting, 2026
- [W3] Xuan, C.[†], **Wang, J.[†]**, Xie, W., Fang, H., Jiang, M., & Lee, E. W. M. Smart Evacuation During Building Fires: A Systematic Scoping Review of Real-time Sensing, Adaptive Decision-Making, and Collaborative Guidance. Under review at *Journal of Building Engineering*
- [W2] Aguirre Garrido, V., **Wang, J.**, Nguyen, T., Marion, J., Pitts, B. J., & Yih, Y. Would you confide in AI? Exploring profile differences among users and non-users of AI for emotional support. Under review at *Behaviour & Information Technology*
- [W1] Ju, S., Wu, M., Wu, Y., **Wang, J.**, Su, Z., Liu, D., Fu, Y., Pu, D., Li, K., Han, Y., Yin, R., & Wang, X. SurgVision: AI-Augmented Spatial-Temporal Scaffolding for Learning Surgical Anatomy from Laparoscopic Videos. Under review at *Anatomical Sciences Education*

RESEARCH GRANT

- [G1] **Principal Investigator** | Chinese National College Student Innovation Training Program
- Topic: The role of assistance behavior in high-rise hospital evacuation
 - Funded by the *Ministry of Education of China* (¥5,000). Project evaluated as **Excellent** upon conclusion

PRESENTATION

- [P1] **Wang, J.**, & Xie, W. (2025, June). The role of assistance behavior in high-rise hospital evacuation: An agent-based simulation. *The 9th Digital Belt and Road Conference*. Chengdu, China. (**Young Scientist Award for Outstanding Oral Presentation**)

TEACHING EXPERIENCE

Teaching Assistant IE 559: Cognitive Engineering of Interactive Software	Purdue University Fall 2026
Teaching Assistant IE 545: Engineering Economic Analysis	Purdue University Fall 2026
Teaching Assistant & Lab Instructor IE 386: Work Analysis and Design	Purdue University Spring 2026

Teaching Assistant	UIP 2025: Multi-hazard Impact and Risk Assessment	Sichuan University Summer 2025
Teaching Assistant	IE 1055: Engineering Management	Sichuan University Spring 2025
Teaching Assistant	ENGR 0012: Engineering Computing	Sichuan University Spring 2025
Teaching Assistant	MATH 0290: Differential Equations	Sichuan University Fall 2024
Teaching Assistant	ENGR 0031: Electric Circuits	Sichuan University Fall 2024
Teaching Assistant	ENGR 0011: Introduction to Engineering Analysis	Sichuan University Spring 2024

LEADERSHIP EXPERIENCE

Mentor	Purdue IE Graduate Mentorship Program	West Lafayette, USA Aug 2026-Jul 2027
Communication Director	HFES Purdue Chapter	West Lafayette, USA Aug 2026-Jul 2027
Communication Specialist	HFES Purdue Chapter	West Lafayette, USA Aug 2025-Jul 2026
President	Industrial Engineering Class 2, SCUPI	Chengdu, China Sep 2021-Jun 2025
President	Disaster Reduction & Emergency Management Association, SCU-IDMR	Chengdu, China Sep 2022-Jun 2024
Captain	Volleyball Team of SCUPI	Chengdu, China Sep 2022-Jun 2024
Peer Leader & Director	SCUPI Peer Advisor Program	Chengdu, China May 2022-Jun 2024
Peer Advisor	SCUPI Peer Advisor Program	Chengdu, China May 2021-Jun 2022

SKILLS

Skills: Python, MATLAB, C++, Minitab, R, SIMUL8, Tableau, SPSS, and GitHub Copilot
Languages: English (TOEFL: 107, Reading: 28, Listening: 28, Speaking: 26 and Writing: 25) and Mandarin (native)